

Yukang Shen

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Research Interests: Physical AI — continual embodied learning, multimodal embodied representation learning, hierarchical control policies, and compact brain-inspired agent architectures

EDUCATION

Kennesaw State University, M.S. Software Engineering **2025.1 – Present**
Graduate Research Assistant

Heilongjiang University, B.Eng. Computer Network Engineering **2016.9 – 2020.7**
National Encouragement Scholarship; First-Class Scholarship (2018, 2019)

HONORS

Second Place, KSU C-Day Computing Showcase, Fall 2025 — Synthetic data generation and medical perception system design.

RESEARCH EXPERIENCE

Real-Calibrated Synthetic-First Data Engine for Medical Perception **2025 – 2026**

- Developed an synthetic-first data engine for deltoid injection-site segmentation, integrating diffusion-based generation, controllable rendering, CLIP filtering, and geometry-guided data selection.
- Built dataset curation, annotation, and pseudo-label refinement tools; evaluated synthetic-to-real transfer using AP, boundary distance, JS divergence, and safety-zone metrics.

Physical AI Infrastructure for Continual Embodied Learning **2026 – Present**

- Developing a biologically inspired memory infrastructure for Physical AI, focusing on modular brain-region-like memory organization, multi-channel feature learning, and continual adaptation from physical interaction.
- Investigating how embodied agents can move beyond demo-centric modular pipelines by integrating perception, memory, safety constraints, and decision signals into a compact lifelong-learning architecture.

WORK EXPERIENCE

City University of Hong Kong, Research Assistant **2024.11 – 2025.1**
Tech: Python, React

- Developed a secure, encrypted data-marketing platform supporting cross-institution research collaboration.
- Implemented frontend dashboards and backend API modules focused on privacy, data integrity, and workflow usability.

SenseTime Group Limited, Software Engineer **2020.11 – 2023.9**
Tech: TypeScript, React, Electron, Node.js, Webpack, CI/CD

- Led frontend development for shared component libraries, internal development tools, OA systems, Enterprise WeChat automation, AR navigation, and enterprise platforms.

PUBLICATIONS

- *A Real-Calibrated Synthetic-First Data Engine*. **Yukang Shen**. Preprint, 2026.
- *6G-enabled Edge AI for Metaverse: Challenges, Methods, and Future Research Directions*. Luyi Chang, Zhe Zhang, Pei Li, Shan Xi, Wei Guo, **Yukang Shen**, Zehui Xiong, Jiawen Kang, Dusit Niyato, Xiuquan Qiao, Yi Wu. *Journal of Communications and Networks*, 2022.

SKILLS

Programming: Python, PyTorch, JavaScript/TypeScript, Node.js, React, Electron

AI Research: Computer Vision, Multimodal Perception, 3D Vision, SLAM, Synthetic Data, Segmentation

Research Systems: Data Engine Design, Dataset Curation, Annotation Pipelines, Model Evaluation, Synthetic-to-Real Calibration